

(a) *Cirsium pannonicum*
purple; upright



(b) *Cirsium mohavense*
pale; inclined



(c) *Cirsium subcoriaceum*
pink; inclined



(d) Why localize?

A full scene may contain leaves, stems, background and several heads.

YOLO fixes one capitulum as the measurement unit. It does not score traits.

(e) Tight crop



(f) Context crop



(g) Corolla mask



(h) Outline + hull



(i) Orientation relative to image vertical



(j) Continuous outputs

Orientation	0.7°	usable
Lightness	56.5	usable
Chroma	33.3	usable
Hue	300.2°	usable
Aspect ratio	1.59	usable
Circularity	0.77	usable
Solidity	0.95	usable

Detailed example: *Cirsium pannonicum*; observation 295734197; photo 532420148; CC BY.

27 registered
continuous endpoints

22 measured
finite v2 outputs

5 unexecuted
remain missing evidence

Detector = crop
does not score traits